

Lab 1 – Redflag

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1 Introduction

Misinformation is spread across many social media platforms across the internet making it difficult for users to find reliable and trustworthy content. For users that typically browse through social media or news sources, there are no apparent signs that the information they come across is biased or misinformation. This leads to them sharing this misinformation as about 33.1% of false headlines are shared from users believing the information was real (Pennycook et al., 2021). One reason for why users are unlikely to look further into the information they see is real is because it takes away from their experience of browsing. Users do not want to have to go in and out of their browsing experience to fact check every option. There are other solutions for this that help with quickly identifying the information, but as stated before these all detract from the original browsing experience and are seen as obtrusive. Another issue with the spread of misinformation is that the users, if they decide to try to look further into misinformation, will be spending more time on just finding good sources of information. It can be hard to judge and differentiate what other sources may also be spreading misinformation as well or if they are biased. There are no real time tools in the current ecosystem of fact checking directly within the browsing experience. This is where Redflag comes in, as it helps with informing the user with various sources of information with no bias towards any particular one. Rather than explicitly pointing out misinformation to the user and popping up with sources of why it may be wrong, it is subtle in its implementation of letting the user know if they may want to look further into the information they see. Redflag also solves the issue of the user needing to find reliable sources for misinformation. Rather than having the user decide on which sources are trustworthy or what bias they may have, Redflag curates these sources and balances them all out so no one source is over-represented.

2 Product Description

- RedFlag is a fact-checking and bias indicator utility that works across multiple platforms. It runs in the background and scans text to find any inaccuracies and provide sources for the user to look into. This information and sources provided give the user a way to look further into the misinformation. There is also the BiasStar which will add a star to any politically divisive content and offers further viewpoints from sources across the political spectrum.
- [Figure 2: RedFlag High Level Concept Diagram]

2.1 Key Product Features and Capabilities

- Real-Time Integrated Detection
 - Feature: Scans the user's screen to find any misinformation.
 - Significance: Works automatically in the background without requiring user input
 - Addresses: Tackles the inconvenience of the user having to leave their current browsing experience by bringing sources directly to them
- Highlight Warnings
 - Feature: Highlights any questionable information on the users' browsing experience with no user input
 - Significance: Does not intrude on the users' experience and presents the highlight subtly
 - Addresses: Attempts to intervene before the user shares misinformation, while not entirely interrupting the experience
- Balanced Perspectives Engine (BiasStar)

- o Feature: Identifies political content and provides multiple viewpoint sources across the political spectrum
 - o Significance: Doesn't label ideologies explicitly but offers an evaluation from every perspective
 - o Addresses: Tackles how social media may reinforce bias and echo chambers
- On Demand Explanations
 - o Feature: Detailed overviews are provided on the flagged information and explained to the user
 - o Significance: Allows the user to make a more information decision without dictating how the user should think on the topic
 - o Addresses: Provides better credibility towards content and allows the user to think further on the information

2.2 Major Components (Hardware/Software)

[Figure 3: Major Functional Component Diagram (MFCD)]

- Presentation Layer (Frontend)
 - o Components: The RedFlag browser extension
 - o Function: Displays real time alerts and a view panel for users
 - o Tech Stack: Built with Javascript for browser extensions
- Application Layer (Backend)
 - o Components: Content detection, Misinformation Analysis, Fact Verification, Balanced Perspectives Engine, and Alert & Recommendation Generator
 - o Tech Stack: Python (Flash/Django) for the information datasets

- Data Layer (Database)
 - Components: PostgreSQL database
 - Stored Data: Source Credibility Database (Includes reliability scores, and political alignments), User Interaction Data, and Content Intelligence Data (Known false claims, topic classifications)
- External Interfaces / APIs
 - Fact checking API: Compares questionable information against the database
 - News API: Takes public news at input and categorizes by topic
 - LLM API: Parses database information to summarize its findings and increase fact-checking speed

3 Identification of Case Study

- For whom is the product being developed? Why? Who else might use this in the future?
- - WHO: 16 year-old teenager, Daiskue, spends large amounts of his free time scrolling through his phone on Instagram
- - PROBLEM: He recognizes some false information on his Instagram feed, but doesn't want to spend the time researching every post he comes across.
- - FEATURES: Redflag misinformation indicators
- - SCENARIO: Daiskue enables Redflag on his browser, and indicators pop up on questionable text. He receives multiple sources of information on questionable topics, and is able to go to those sources immediately.
- - FUTURE: Word of mouth, discussion of Redflag with his friends

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4 Glossary

- Confirmation Bias
 - The tendency to interpret new information and evidence as confirmation of one's existing beliefs or theories, often while ignoring or discounting opposing evidence.
- Credibility Score
 - A score given to a claim that is found online rating the credibility of the claim, or the likelihood that the claim is correct.
- Friction
 - Something that causes an application to not be seamless while working.
- L.L.M.
 - A Large Language Model (also known as a ChatBot) such as ChatGPT, Gemini, or Claude.
- Misinformation
 - Provably false or inaccurate information.
- Disinformation
 - False or inaccurate information that is shared with the intention to mislead others.
- End-user

- o The person who ultimately uses or is intended to use a computer system or software application.
- Troll
 - o A person who posts inflammatory, insincere, creating discord, or off-topic messages in an online community.

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5 References

Pennycook, G., Epstein, Z., Mosleh, M., Arechar, A. A., Eckles, D., & Rand, D. G. (2021). Shifting Attention to Accuracy Can Reduce Misinformation Online. *Nature*, 592(7855). <https://doi.org/10.1038/s41586-021-03344-2>